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ACADEMIC NETWORK FOR A GREEN & INNOVATIVE EUROPE

WP2–Setting-up the Graduate Study Program (GSP) "Transition to Green for Sustainable Society"

Academic Network for a Green and Innovative Europe

D2.1.2- List of topics to use in curriculum

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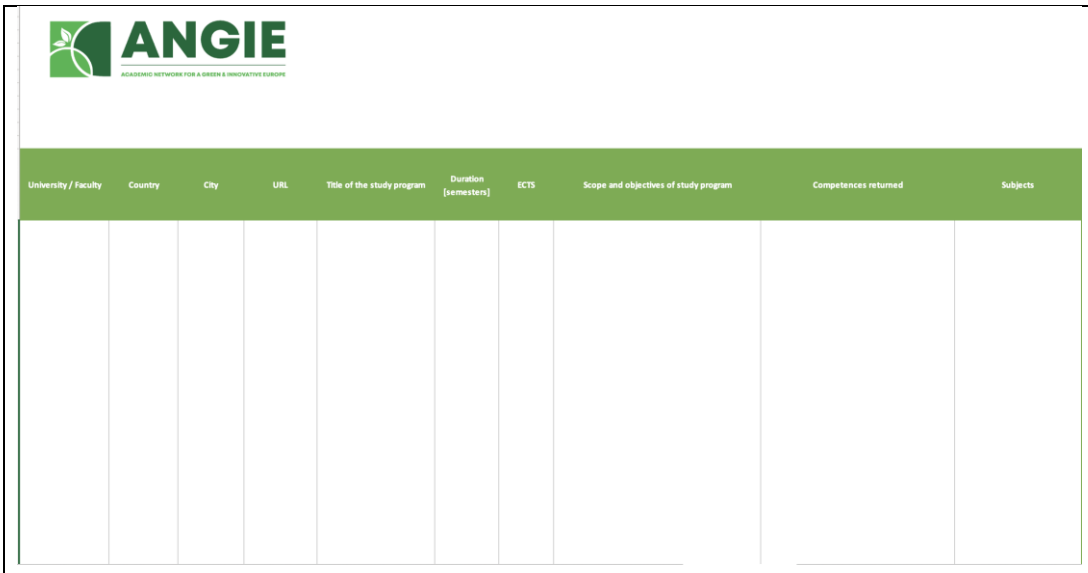
1 Foreword

This deliverable aim to create a list of topics to be used in the Master curriculum.
To reach that goal an inventory and analysis of WATERIC topics will be presented, followed by the selection of the most appropriate topics, according to the partners inventory and the problems identified in the Deliverable D.2.1.1

The topics selected had in consideration the potential specific measures to be taken when implement the transition to green and the heterogeneity of the learning audience.

2 Overview of the methodology to select the most relevant Topics

To obtain a list with the most relevant learning Topics or Disciplines the following methodology was used:
First, spreadsheets, as the ones represented in figure 1 and figure 2, were prepared and distributed among each partner to collect existing masters and respective disciplines, per country. After collecting all the information, a list of topics/disciplines were created by Domain: Agriculture, transportation, Waste, Energy production, Industrial activities and Residential activities, which can be observed in section 3.



University / Faculty	Country	City	URL	Title of the study program	Duration [semesters]	ECTS	Scope and objectives of study program	Competences returned	Subjects

a)

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University / Faculty	Country	City	URL	Title of the study program	Duration (semesters)	ECTS	Scope and objectives of study program	Competences returned	Subjects
ENAIDE Business School	Spain	On-Line		Master's Degree in Water Technology and Management	6 months	60	• Expand knowledge about water management and treatment. • Identify the processes used in the treatment of drinking water. • Acquire fundamental knowledge about waste management. • Know the necessary resources and the planning required to implement distribution and sanitation facilities. • Describe the essential requirements to establish an adequate environmental management system.	This Official University Master's Degree in Water and Waste Technology and Management not only addresses water management, but also waste management. The Master focuses on understanding the associated risks and the necessary preventive and corrective measures. It offers to the student training in environmental management and auditing according to the ISO 14001 standard, to comply with current laws. It will allow the student to acquire the necessary knowledge to comprehensively address the management and treatment of water and waste. And it will provide to the student with an overview of the direction and management of a water company. Students will learn the procedures necessary to implement a management system in accordance with the ISO 14001:2015 management and audit regulations.	Module 1: quality and contamination of water Module 2: water management and treatment, etc Module 3: water supply and distribution Module 4: water treatment projects Module 5: management and management of water companies Module 6: waste management Module 7: soil contamination and recovery of degraded spaces Module 8: environmental management and auditing (iso 14001 - iso 19011) Module 9: final master project
ENAIDE Business School	Spain	On-Line		Official University Master's Degree in Management of Water Purification and Treatment Stations ETAP, WWTP, Soil and Waste	6 months	60	Get up-to-date training on purification and treatment. Study the use of advanced technologies and innovation. Analyze the environmental and economic impacts. Apply the knowledge gained in practical situations	The Official University Master in Management of Water Treatment Plants and Water Treatment ETAP, EDAR, Suelos and Residuos + 60 Credits has been designed to address the concepts, types, methods and treatment of the three previously mentioned areas (Water, Suelo and Residuos) , toasting a complete and updated education so that students can develop professionally in this field.	Module 1: Water management and treatment ETAP and EDAR Module 2: Quality and contamination of water Module 3: Water collection and distribution Module 4: Water treatment projects Module 5: Management and control of contamination of the environment Module 6: Management of residues Module 7: Contamination of soil and recovery of degraded spaces Module 8: Environmental audit Module 9: Final master project
Miguel Hernández University	Spain	Hybrid	https://masteres.iduos.umh.es	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	12 months	60	Analyze and solve scientific and technological problems in the context of residues Please advise waste managers Plan strategically and valorize the residuals Check the quality of the fertilizer products and refinements Manage waste installations Desarrollar analysis of economic and environmental viability of waste management	General Competencies Acquisition of scientific knowledge in organic waste management. Identification and resolution of problems related to organic waste management. Design and project management skills. Ability to adapt to scientific advances, continuous improvement, innovation and creativity in the field of organic waste management. Ability to analyze and synthesize information. Critical and analytical capacity in the environmental field. Capacity for decision-making. Ability to communicate oral and written knowledge acquired. Ability to work in multidisciplinary and multicultural teams. Ethical and environmental commitment to the development of sustainable scientific solutions with the human and natural environment. Ability to manage databases and other tools available to obtain Scientific information. Specific Competencies Ability to design experiments related to waste management and data analysis. Knowledge about the different economic resources for financing research. Identify current organic waste management techniques and the sectors with the greatest production of them. Manage current European and Spanish legislation on organic waste. Classify the different types of treatment and stabilization of organic waste and know their basis. Know the characteristics, treatment methods and recovery of the main types of organic waste generated in different sectors. Learn different practical cases of organic waste management.	Fundamentals of treatment and valorization of organical residuals 3 mandatory herramientas metodológicas para la investigación 6 mandatory Introduction to the management of organic residues and applicable regulation 3 mandatory Practical approach to anaerobic digestion 3 optional Physicochemical, biochemical and microbiological aspects of the composting process. Eval. Quality 4.5 optional Technical aspects in the development and control of the composting process 3 optional Anaerobic digestion. Fundamentals, influencing factors and available technology 6 optional Emission of gases and odors in organic waste management: control and attenuation 3 optional Fundamentals of the composting and vermicomposting process 3 optional Environmental, economic and energy implications of organic waste management 4.5 optional MSW and sludge valorization 3 optional Itinerary 2 Agronomic application of treated and stabilized organic waste 3 optional Application effect Of organic residues treated and stabilized. About prop. Physic., chem. And biol. From the ground 4.5 optional Evaluation of pathogen suppressiveness of waste and compost 3 optional Organic agriculture as a recipient of treated and stabilized organic waste 3 optional Handling, dosage and agronomic management of treated and stabilized organic waste 4.5 optional

b)

Figure 1. Spreadsheets distributed among each partner to identify existing masters in the WATERIC field, a) Sheet sent to the partners; b) Example of a completed sheet by Malaga University

Title of study program	Subjects	ECTS

PAY ATTENTION!

A year of full-time studies at university level is generally worth 60 ECTS credits, and defined as equal to 1,500 - 1,800 hours of study work. This means **1 ECTS is equal to between 25 to 30 hours** (with the UK being one exception). The exact number of hours is different from country to country.

A few examples:
United Kingdom: 60 ECTS = 1,200 study hours → 1 ECTS = 20 study hours
Austria, **Ireland**, **Italy**, **Malta**: 60 ECTS = 1,500 study hours → 1 ECTS = 25 study hours
Finland, **Lithuania**, **Sweden**: 60 ECTS = 1,600 study hours → 1 ECTS = 27 study hours
Netherlands, **Portugal**: 60 ECTS = 1,680 study hours → 1 ECTS = 28 study hours
Germany: 60 ECTS = 1,800 study hours → 1 ECTS = 30 study hours

a)

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	A	B	G	H
1	Title of study program	Subjects	ECTS	
2	Master's Degree in Water Technology and Management	Module 1: quality and contamination of water	6	
3	Master's Degree in Water Technology and Management	Module 2: water management and treatment, etc	6	
4	Master's Degree in Water Technology and Management	Module 3: water supply and distribution	6	
5	Master's Degree in Water Technology and Management	Module 4: water treatment projects	6	
5	Master's Degree in Water Technology and Management	Module 5: management and management of water companies	6	
7	Master's Degree in Water Technology and Management	Module 6: waste management	6	
3	Master's Degree in Water Technology and Management	Module 7: soil contamination and recovery of degraded spaces	6	
9	Master's Degree in Water Technology and Management	Module 8: environmental management and auditing (iso 14001 - iso 19011)	6	
0	Master's Degree in Water Technology and Management	Module 9: final master project	12	
1	Official University Master's Degree in Management of Water Purification and Treatment S	Module 1: Water management and treatment ETAP and EDAR	6	
2	Official University Master's Degree in Management of Water Purification and Treatment S	Module 2: Quality and contamination of water	6	
3	Official University Master's Degree in Management of Water Purification and Treatment S	Module 3: Water collection and distribution	6	
4	Official University Master's Degree in Management of Water Purification and Treatment S	Module 4: Water treatment projects	6	
5	Official University Master's Degree in Management of Water Purification and Treatment S	Module 5: Management and control of contamination of the environment	6	
6	Official University Master's Degree in Management of Water Purification and Treatment S	Module 6: Management of residues	6	
7	Official University Master's Degree in Management of Water Purification and Treatment S	Module 7: Contamination of soil and recovery of degraded spaces	6	
8	Official University Master's Degree in Management of Water Purification and Treatment S	Module 8: Environmental audit	6	
9	Official University Master's Degree in Management of Water Purification and Treatment S	Module 9: Final master project	12	
0	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Fundamentals of treatment and stabilization of organic waste 3 mandatory	3	
1	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Methodological tools for mandatory research 6	6	
2	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Introduction to organic waste management and applicable regulations 3 mandatory	3	
3	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Organic waste of agroindustrial and urban origin 3 mandatory	3	
4	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Organic waste of agricultural and livestock origin 3 mandatory	3	
5	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Master's thesis 12 master's thesis		
6	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Itinerary 1		
7	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Practical approach to anaerobic digestion 3 optional	3	
8	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Physicochemical, biochemical and microbiological aspects of the composting process. Eval. Quality 4.5 option	4.5	
9	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Technical aspects in the development and control of the composting process 3 optional	3	
0	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Anaerobic digestion. Fundamentals, influencing factors and available technology 6 optional	6	
1	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Emission of gases and odors in organic waste management: control and attenuation 3 optional	3	
2	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Fundamentals of the composting and vermicomposting process 3 optional	3	
3	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Environmental, economic and energy implications of organic waste management 4.5 optional	4.5	
4	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	MSW and sludge valorization 3 optional	3	
5	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Itinerary 2		
6	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Agronomic application of treated and stabilized organic waste 3 optional	3	
7	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Application effect Of organic residues treated and stabilized. About prop. Physic., chem. And biol. From the gro	4.5	
8	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Evaluation of pathogen suppressiveness of waste and compost 3 optional	3	
9	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Organic agriculture as a recipient of treated and stabilized organic waste 3 optional	3	
0	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Handling, dosage and agronomic management of treated and stabilized organic waste 4.5 optional	4.5	
1	Master's Degree in Research in Management, Treatment and Recovery of Organic Waste	Obtaining high added value materials through composting 3 optional	3	

b)

Figure 2. Spreadsheets distributed among each partner to identify existing Disciplines/Topics; a) Sheet sent to the partners; b) Example of a completed sheet by Malaga University

After gathering all partners worksheet information and analysing in combination with the most relevant problems and needs identified in Deliverable D.2.1.1, in the different 7 sectors, Agriculture, waste, transport, energy production, residential activities and industrial activities, and in the 8 partners countries, namely Romania, Austria, Spain, Italy, Slovenia, Slovakia, Germany and Portugal, a selectin of subjects was obtained.

3 Summary of the WATERIC topics to be chosen by the partners

Based on the Excel spreadsheet, represented in figure 2. b), which was filled by the partners, considering the result of disciplines inventory from the Masters of the different countries, topics were gathered by domain, namely: Construction; Industrial activities; Transportation; Energy production, Waste management; Residential activities and Agriculture. Additionally, from the deliverable D.2.1.1, emergent topics were also highlighted and listed in this report. These lists will allow the ShakeAM partners to select the most relevant topics for the curriculum construction.

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3.1 List of Topics by Domain

In this section a list of topics by field is presented.

3.1.1 Construction field topics

Integrated environmental valorisation of construction materials
Eco-materials and sustainable construction
Urban planning
Construction management
Energy management in buildings
Environmental management in building
Occupational and safety management in construction
Buildings/construction sustainability
Quantification, minimization and recycling. Building with recycled materials
Urban development
Renewable materials
Sustainable heating, ventilation and air conditioning (HVAC)
Smart building controls
Climate friendly construction
Aspects of sustainable urbanism
Interactions between building materials
Sustainable lighting technology
Ecological rehabilitation / reconstruction
Infrastructure and urban climate
Innovative buildings & urban development
Profitability analysis & building law
Comfort & acoustics
Heating & air conditioning technology
Ventilation & sanitation
Electrical installation technology
Building automation & user integration
Increasing efficiency & cost-effectiveness
Smart cities
Fundamentals of urban infrastructure systems
Fundamentals of urban planning
Spatial planning
Revitalization, local image, and architectural heritage protection
Wood composites
Renewable and wood-based materials in construction

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Energy refurbishment of buildings
Ergonomics and built environment
Recycled waste in construction

3.1.2 List of Construction topics obtained From D2.1.1

Energy efficiency of the buildings
Sustainable materials
Sustainable urban planning
thermal performance of the buildings
Climate neutral new buildings
Waste from construction

3.1.3 Industrial activities field topics

Livestock waste management
Waste gas treatment systems
Wastewater treatment technologies
Industrial pollution prevention
Characterization, management and treatment of water pollution
Life cycle of products and services
Energy efficiency in industrial processes
Alternative energy in industrial processes
Renewable materials
Clean technology/eco-technology
Rational use of materials
Resource recovery and recycling technologies
Renewable energy transition
Renewable power, heating and cooling
Industrial hygiene and safety
Biodegradable packaging and polymer
Biological treatment of industrial waste
Recycling management of by-products
Environmental technologies
Eco-toxicology
Chemical regulation
Human safety

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Safe innovation & green product design
Advanced packaging technology
Packaging testing systems and technology
Sustainability and resource management
Trends and future markets in packaging
Green logistics

3.1.4 List of Industrial activities topics obtained From D2.1.1

Greenhouse gases emissions
Resource efficiency/circular economy
Environmental performance index (EPI) and environmental compliance
Energy efficiency in industry
Waste from manufacturing
Raw material consumption (RMC)
Green industrial policy
Air emissions in industry
Circular material use

3.1.5 Transports field topics

Management of atmospheric pollution
Management of noise pollution
Sustainable mobilities – theories, concepts, approaches
Mobility solutions
Basic knowledge for the mobility transition
Urban mobilities
Managing and recycling of wreck vehicles
Advanced technics for vehicle evaluation and pollution mitigation
Present and possible systems for vehicle engines
Transport routes and their connection
Tourism economics and the impact upon the environment
Transportation planning & design
Transportation choices & technology
Sustainable transportation from a systems perspective
Health & sustainable transportation
Environmental science and sustainability for transport
Shaping future transport systems
Public transport planning and management

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Traffic management
Analysing transport and society
Transport and urban pollution
Energy use in vehicles
High-voltage technology in vehicles
Hybrid drive technology
Mechanical drive components
Automotive IT-Security
Charging infrastructure
Charging technology
Lightweight construction
Modularisation in automotive engineering
Ecological aspects of mobility
Process and quality management in the automotive sector
Energy in maritime transport
Traffic and sustainable development

3.1.6 List of Transports field topics obtain From D2.1.1

Emissions of fuels combustion/greenhouse gas emissions
Inadequate transportation infrastructure
Alternative fuels/sustainable transportation
Batteries
Advanced biofuels
Modal shift in passenger and freight transport to environmentally friendly transport modes
Increase fuel efficiency in road transport
Increase the share of clean energy sources in road transport
Alternative fuels infrastructure regulation
Energy consumption in transport by type of fuel
Policies and measures by transport sector

3.1.7 Waste field topics

Organic waste management
Emission of gases and odours in organic waste management
Organic waste of agro-industrial origin

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Introduction to organic waste management and applicable regulations
Environmental, economic and energy implications of organic waste management
Material cycles
Waste characterization: preparation for reuse and recycling
Technologies for resource recovery: recycling and recovery
Tools for circular economy
Industrial waste water treatment
Environmental impact of materials degradation
Waste and hazardous substances
Eco-toxicology
Exterior deposits. Geological risks
Solid waste management
Residual gases treatment and purification. Air quality monitoring
Waste prevention and preservation of resources in packaging
Textile waste management
Material and energy waste recovery

Note:
Organic wastes include animal manures, crop residues, food processing wastes, municipal biosolids and wastes from some industries. Organic wastes are typically by-products of farming, industrial or municipal activities, and are usually called “wastes” because they are not the primary product.

Livestock industries produce meat, milk and egg. Livestock Waste" means livestock excreta, bedding material, rain or other water, soil, hair, feathers or other debris normally included in animal waste handling operations. Improperly managed livestock wastes can lead to faecal contamination of waters receiving agricultural runoff. The danger to humans lies in the possibility of these faecal organisms entering water and food supplies

3.1.8 List of Waste field topics obtained From D2.1.1

Improper waste disposal and landfill use
Plastic pollution
Circular economy including: • Packaging and plastics •Sustainable textiles •Electronics and ICT •Construction and buildings •Batteries and vehicles •The food chain. •Critical raw materials • Repairing and reusing goods
Source of waste generation
waste treatment

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Waste management: • Recovery-recycling and backfilling (excluding energy recovery) • Energy recovery • Recovery - backfilling • Recovery - recycling • Incineration (without energy recovery) • Disposal - landfill • Disposal - other • Landfill / disposal

3.1.9 Energy production field topics

Energy management
Energy, natural resources and climate change
Fuel cells and hydrogen economy
Renewable energies
Energetic use of renewable materials
Principles of sustainability/Design of sustainable products & services
Smart grids
Power system economics
Power system operation and stability
Planning & Operation of distributed energy sources
Sustainable fuels for the energy transition
Automation of complex power systems
Advanced materials for renewable energy
Energy and environment
Energy storage
Control technology
Environment, energy and society
Digital systems in the energy industry
Supply and demand management
Efficient decentralised energy technologies
Renewable heating and cooling
Bioenergy: plant technology and project planning
Heat pumps: plant technology and project planning
Electrical energy technology
Market intelligence in the energy sector
Energy networks
Fuels for sustainable development

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3.1.10 List of Energy production field topics obtained from D2.1.1

Renewable energy
Energy storage
Grid modernization
Energy efficiency
Energy security
Energy production: • crude oil, • petroleum products, • natural gas and manufactures gases, • electricity and derived heat, • solid fossil fuels, • renewables

3.1.11 Residential activities field topics

Organic waste of urban origin
Urban metabolism and ecological urbanism
Environmental impact of plastics
Sanitary engineering in developing countries
Renewable power, heating and cooling
Sustainable residue, waste and water processing technologies
Protection of the population against highly-toxic chemical and biological agent
Environmental impact of materials degradation
Ecological rehabilitation of degraded areas
Generation and sustainable usage of energy
Ergonomics and ecology of work environment
Municipal energy management
Drinking water supply and treatment

3.1.12 List of Residential activities field topics obtained From D2.1.1

Energy Consumption (heat pumps, appliances and lightening systems,...)
Waste production/Reduction
Water waste/Conservation
Reduction in the consumption
Waste from households
Waste from textiles
Municipal waste

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Responsible consumption

3.1.13 Agriculture field topics

Water management and treatments
Soil contamination and recovery of degraded spaces
Agriculture waste management
Contamination of soil and recovery of degraded spaces
Obtaining high added value materials through composting
Land management
Management of contaminated soils and sediments
Conservation of biodiversity
Socioeconomics of rural development and food security
Sustainable international agriculture: basic principles and approaches
Ecotoxicology of plant protection
Areas of conflict between agriculture and the environment
Forest management under different climatic conditions
Contemporary temperate forest and nature management
International forest policy and economics
Landscape ecology and agroecology
Global environmental problems and international nature conservation
Sustainable land management
Insects for food and feed production systems
Conservation of nature and biodiversity
Strategies and methods for biodiversity conservation
Fauna analysis and monitoring
Land degradation by soil destruction - improvement measures
Landfills stability
Protected areas and adequate assessment
Soils protection and decontamination
Modern techniques for soil protection
Plant ecology
Insect biology
Drainage and irrigation
Vegetation ecology
Soil ecology
Ex situ plant conservation
Organic agriculture
Soil pollution

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Floral biology and pollination ecology

3.1.14 List of Agriculture field topics obtained from D2.1.1

Pesticide use
Water consumption/withdrawals in agriculture
Agriculture sustainability
Gas emissions from agriculture
Soil loss by water erosion
Energy/resources consumption

3.2 Topics Common to all domains

Topics common to all domains were identified and listed in the table below

European Environmental Law and Politics
Ethics and Integrity
Soft Skills for Green transition
Integrated environmental, quality and safety management
Environmental risk management
Digitalization
Climate Risks

4 Disciplines selected for the Master Curriculum

The analysis of the information gathered and crossing the topics obtained from the Excel spreadsheets, filled in by the partners, and those obtained from D2.1.1, and considering that are topics that are common in different domains, it was possible to define 14 disciplines/subjects, distributed according with table 1.

Table 1- Number of disciplines per domain

Domains	Nr. of Disciplines
Construction-Industrial activities	5
Transportation and Logistics	5
Energy production	2
Waste management	2
Residential activities	1

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Agriculture	1
Transversal Topics	5

In the next subsection the 12 disciplines are presented. The topics from each discipline can also be observed, those are also the results of the analysis made in this activity.

4.1 Construction-Industrial activities

Subject #1	Title: Critical Raw Materials
	Identification and assessment of critical raw materials
	Sustainable extraction and processing techniques
	Substitution and recycling of critical raw materials
	Supply chain management and security in raw materials
	Policy and regulatory frameworks

Subject #2	Title: Resources Management
	Foundations of sustainable resource management
	Life Cycle Assessment (LCA) and Environmental Impact Analysis (EIA)
	Circular economy in engineering
	Sustainable energy and resource use
	Economic and policy aspects of sustainability
	Risk management and climate change adaptation
	Systems thinking and modelling for resource management
	Technological innovations for sustainable engineering
	Ethical and social dimensions of sustainability
	Practical applications and case studies

Subject #3	Title: Materials in Circular Economy
	Introduction to Circular Economy Concepts
	Material Properties and Their Impact on Circularity
	Design Strategies for Circular Products
	Material Flow Analysis and Waste Management
	Sustainable Material Innovation
	Advanced Manufacturing for the Circular Economy
	Regulations, Standards, and Compliance

Subject #4	Title: Green Buildings (nZEB)
	Green Building Standards
	Near Zero Emission Buildings
	Energy Performance
	Water Conservation
	Indoor Environmental Quality

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Subject #5	Title: LCA and sustainable rehabilitation of buildings
	Life cycle Analysis (LCA)
	Sustainable materials and technologies for building rehabilitation
	Energy-efficient building systems
	Water management in rehabilitated buildings
	Indoor environmental quality (IEQ) improvement
	Sustainable Building Operations and Maintenance

4.2 Transportation and Logistics

Subject #1	Title: Sustainable Supply Chain Management
	Integration of Circular Economy principles in Supply Chain Management
	Green procurement and supplier collaboration
	Carbon footprint reduction in Supply Chain operations
	Ethical and social responsibility in Supply Chains
	Technology and innovation for sustainable Supply Chains
	Integration of Circular Economy principles in Supply Chain Management

Subject #2	Title: Sustainable Transportation
	Emissions of fuels combustion/greenhouse gas emissions
	Modal shift in passenger and freight transport to environmentally friendly transport modes
	Transportation choice & technology
	Multi/Intermodal transport
	Innovative transport concepts
	Policies and measures by transport sector

Subject #3	Title Warehousing and Distribution Centres
	Energy Consumption and Efficiency in Warehouses
	Sustainable Waste Management Practices
	Green Building Practices and Certifications
	Automation and IoT for Sustainable Warehousing
	Eco-friendly Materials and Packaging Solutions

Subject #5	Title: Optimization in Warehousing
	Introduction to Warehouse Optimization
	Warehouse Layout and Design Optimization
	Inventory Management Optimization
	Data Analytics in Warehousing
	Technology and Automation in Warehousing
	Performance Metrics and KPIs
	Continuous Improvement Strategies
	Case Studies in Warehouse Optimization

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4.3 Energy production

Subject #1	Title: Sustainable Energy Systems and Environmental Impact
	Overview of energy production systems: fossil fuels, nuclear, and renewables
	Life cycle analysis of energy systems
	Greenhouse gas emissions and climate change
	Air, water, and soil pollution from energy production
	Renewable energy technologies (solar, wind, hydro, and biomass)
	Sustainable energy policies and regulations
	Case studies on successful sustainable energy projects

Subject #2	Title: Climate Change Mitigation and Adaptation in the Energy Sector
	Climate science and the role of the energy sector
	Greenhouse gas inventories and accounting
	International climate agreements and national policies
	Renewable energy integration and grid management
	Energy efficiency measures and technologies
	Carbon capture, utilization, and storage (CCUS)
	Climate adaptation strategies for the energy infrastructure
	Environmental Risk Assessment and Management in Energy Production

4.4 Waste

Subject #1	Title: Waste Management
	Waste classification
	Municipal waste
	Industrial solid waste
	Hazard waste
	Waste water
	Waste to energy
	Waste regulations and compliance

Subject #2	Title: Waste Technologies
	Waste water processing technology
	Solid waste engineering and management
	Air quality control technologies
	Energy technologies for solving environmental challenges
	Sustainable production processes
	Introduction to circular business models

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4.5 Residential activities

Subject #1	Title: Sustainable residential activities
	Domestic energy production and consumption
	Domestic water usage
	Food consumption
	Household products
	Building and renovation
	Electronic use

4.6 Agriculture

Subject #1	Title: Sustainable agriculture
	Soil degradation and erosion
	Greenhouse gas emissions from agricultural practices
	Water scarcity and management
	Impact of pesticides and fertilizers on ecosystems
	Deforestation and land use change
	Monitorization of parameters
	Strategic measures

4.7 Traversal Topics

Subject #1	Title: Soft Skills for Green transition
	Communication in greening process
	Communication and promotion of green transition

Subject #2	Title: Ethics and Integrity
	Research ethics
	Ethics in transition to green

Subject #3	Title: Digitalization
	IT in green transition
	Digital infrastructure for greening process

Subject #4	Title: Research #1
	Industrial practice #1
	Greening case studies analysis #1

Subject #5	Title: Research #2
	Industrial practice #2
	Greening case studies analysis #2

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5 Conclusions

Based on the analysis of the information gathered by the consortium partners it was possible to define 15 relevant Disciplines that will be part of the new Master curriculum. Those disciplines were presented with the respective subtopics associated.
With the disciplines define it will be possible to stablish students learning outcomes and competences, as well as the structure of the master program, which is part of the deliverable D.2.2 and D.2.3.

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